

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

$$f(x) = 4,000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population x years after **2012**, where $x \leq 7$. Which of the following is the best interpretation of **4,000** in this context?

Answer

- A. The estimated initial beetle population for this species and area in **2012**
- B. The estimated beetle population for this species and area **7** years after **2012**
- C. The estimated percent decrease in the beetle population for this species and area each year after **2012**
- D. The estimated percent decrease in the beetle population for this species and area every **7** years after **2012**

Correct Answer: A

Rationale

Choice A is correct. For an exponential function in the form $f(x) = a(b)^x$, where a and b are positive constants and $b < 1$, the initial value of $f(x)$, or the value of $f(x)$ when $x = 0$, is a and the value of $f(x)$ decreases by $100(1 - b)\%$ each time x increases by 1. Therefore, the initial value of the function $f(x) = 4,000(0.75)^x$, or the value of $f(x)$ when $x = 0$, is **4,000**. Therefore, the best interpretation of **4,000** in this context is the estimated initial beetle population for this species and area in **2012**.

Choice B is incorrect. The estimated beetle population for this species and area **7** years after **2012** is $4,000(0.75)^7$, or approximately **534**, not **4,000**.

Choice C is incorrect. The estimated percent decrease in the beetle population for this species and area each year after **2012** is $100(1 - 0.75)$, or **25**, not **4,000**.

Choice D is incorrect. The estimated percent decrease in the beetle population for this species and area every **7** years after **2012** is $100(1 - 0.75^7)$, or approximately **87**, not **4,000**.